

Grade 7 Accelerated
Unit 1
Lesson 5 Homework

Name _____
Date _____
Period _____

Rewrite each expression by applying the distributive property for questions 1-2.

1. $6(x + 4)$

2. $(b - 9)(-4)$

Equations can be solved in more than one way. When an equation is written in the form $a(b + c) = d$, it can be solved by either distributing a first or by dividing both sides the equation by a first. The equation $4(1 - 3x) = 52$ can be solved using either method.

3. Solve the equation $4(1 - 3x) = 52$ using each method.

Method 1 Apply the Distributive Property first	Method 2 Divide both sides by the factor being distributed first
$4(1 - 3x) = 52$	$4(1 - 3x) = 52$

Solve each equation using your preferred method. Then, verify that your solution is correct using substitution.

4. $\frac{3}{10}(x + 2) = 12$

5. $0.2(r - 3) = -10$

6. $\frac{3}{4}\left(x + 5\frac{1}{2}\right) = \frac{5}{8}$

7. Solve the equation $911.32 = 21.74p + 780.88$.

A local plumber charges a \$75 fee for a home visit and \$66.24 per hour while they work. If the total bill was \$339.96, how many hours did the plumber work?

8. Write the equation that represents this situation using the variable h for the unknown quantity.

9. Solve the equation.

10. Explain what the solution means in this context.